Golden Gate Assembly

PROTOCOL

1. For each reaction you set up, **top-label a PCR tube** with the name(s) indicated on the labsheet.
2. Retrieve **water** (white **W** tubes) and **T4 ligase buffer** aliquots (**red** tubes) from the **enzyme freezer** and thaw them at room temperature.
3. Set up the reaction, in order:

6	water
1	10× T4 DNA ligase buffer
2	DNA
0.5	T4 DNA ligase
0.5	restriction enzyme
10	**μL total**

   – If you have **more than 2 fragments** to join, premix **equal volumes of each DNA** in a tube, then use **2 μL of that mix** for the reaction.

   – Be sure you are using the **right one** of BsaI, BsmBI, BseRI, AarI, SapI, BbsI, as indicated in your **construction file and labsheets**.

4. Retrieve the **enzyme cooler**, and add **0.5 μL each** of the ligase and the restriction enzyme to each sample.
5. **Cap, mix, spin.**
6. Run the **GG1** program on the thermocycler.